

Swappable Senses, Stable Contracts: A Modular Platform for Real-Time Adaptive Reading Experiments

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ABSTRACT

Real-time gaze-driven text adaptation has been demonstrated repeatedly, yet studying which adaptations actually help readers remains expensive, because existing prototypes couple sensing, decision logic, and the reading interface into single-purpose codebases. We present a researcher-operated platform for adaptive reading experiments in which sensing, eye-movement analysis, decision strategy, and intervention execution are independently replaceable modules behind stable contracts. A participant reads reflowable text that adapts to their gaze while a researcher mirrors the session live and applies, approves, or delegates typographic interventions; every gaze sample, decision, and intervention lands in a replayable session record. Across eight sessions on research-grade eye-tracking hardware the platform sustained the tracker's rated 90 Hz, kept every measured client round trip within a 100 ms budget, and had its sessions analysed live by an independent team's reading-difficulty pipeline connected through the platform's provider contract without modifying the core. Because the platform measures the positional cost of every intervention it applies, it exposed that our original context-preserving restore moved readers roughly three times further than the reflow it compensated; a revised restore verified over a 66-trial deterministic sweep brought the median displacement below the no-preservation baseline. We further demonstrate sensing replaceability with a work-in-progress webcam module that derives facial state from commodity cameras behind the same contract, sketching a path from laboratory eye tracking toward low-cost deployment. The platform and all session data are open source.

CCS CONCEPTS

• Human-centered computing → User centered design.

KEYWORDS

eye tracking, adaptive reading, gaze-contingent interfaces, modular architecture, context preservation, webcam sensing, research platforms

1 INTRODUCTION

Reading is the interface through which most knowledge work happens, yet for a large population it is a daily struggle. The World Health Organization estimates that at least 2.2 billion people live with some form of vision impairment [26], and age-related macular degeneration alone is projected to affect 288 million people by 2040 [1]. Digital text removes the fixed page: font, size, spacing, and layout can change while a person reads. Combined with eye tracking, this enables a compelling loop: sense where and how a

person reads, detect signs of difficulty, and adapt the typography in the moment [6]. Prior work has repeatedly demonstrated that this loop is feasible [4, 5, 9, 10].

Feasibility, however, is no longer the bottleneck. The open questions are comparative: which typographic interventions help which readers, decided by which strategy, delivered at which moment, and at what cost to the reader's place in the text. Answering such questions requires running many controlled experiments, and this is where current tooling fails. Research prototypes in this space couple the sensor, the analysis logic, the decision policy, and the reading interface into one codebase built for one study [14, 16, 20]. Comparing two decision strategies, or swapping an eye tracker for a cheaper sensor, means editing core code or forking the previous system. Commercial and scientific tools cover parts of the pipeline, but none we surveyed combines real-time gaze sensing, automatic typographic adaptation, and researcher-facing experiment control in one platform (Section 2).

A second, subtler problem compounds the first. An adaptation applied at the wrong moment, or too abruptly, can cost readers their place in the text, and this disruption is usually assumed away rather than measured. Jensen et al. [10] showed that recovery after an intervention is a real cost, measurable as reading resume time, and that intervention design strongly affects it. A platform for adaptive reading experiments should therefore not only preserve the reader's context across interventions but also instrument its own disruption, so that the cost of every applied change appears in the data.

We present a researcher-operated platform for real-time adaptive reading experiments, built within the Reading the Reader project at DTU Compute, that treats these two problems as architectural requirements. Sensing, eye-movement analysis, decision strategy, and intervention execution are four independently replaceable modules behind stable contracts. A participant reads reflowable text on one screen while a researcher mirrors the live session on another, watching gaze and fixations in real time and applying, approving, or delegating typographic interventions. Out-of-process providers, including machine-learning pipelines written by other teams, plug into the analysis and decision seams over a documented protocol and can be exchanged without touching the core runtime. Every gaze sample, oculomotor event, decision, and intervention is captured in a single schema-versioned session record that supports deterministic replay.

We evaluate the platform on research-grade eye-tracking hardware in eight recorded reading sessions (four participants, two conditions), plus a deterministic 66-trial displacement sweep. The platform sustained the tracker's rated 90 Hz sampling with high gaze validity, kept every measured client round trip within its 100 ms

latency budget, and had all eight sessions analysed live by an independent team’s reading-difficulty pipeline connected through the provider contract. The self-instrumentation earned its place: it revealed that our original context-preserving restore over-corrected, moving readers roughly three times further than the reflow it compensated, and it let us verify a revised restore that brings median displacement below the no-preservation baseline.

Finally, we report on a work-in-progress webcam sensing module that exercises the sensing boundary with a signal quite unlike the eye tracker: facial state derived from a commodity camera. It is implemented end to end behind the same contracts, records into the same session format, and yields a coarse reading-difficulty signal, but it has not yet been evaluated with participants; we describe it as a demonstration of sensing replaceability and a path toward low-cost deployment (Section 4).

This paper contributes:

- a modular, researcher-operated platform for real-time adaptive reading experiments, in which sensing, analysis, decision, and intervention modules are independently replaceable behind stable contracts, with reproducible, replayable session records;
- a context-preserving intervention mechanism that measures the positional cost of every change it applies, together with the empirical finding that a plausible restore strategy over-corrected threefold, and a revised strategy verified over a 66-trial deterministic sweep;
- a feasibility evaluation on real eye-tracking hardware covering sensing quality, end-to-end latency against an explicit budget, and live third-party provider integration; and
- a work-in-progress webcam facial-state module demonstrating that the sensing contract extends beyond research-grade eye tracking toward commodity hardware.

2 RELATED WORK

2.1 Eye Movements as a Reading Signal

Decades of reading research establish that oculomotor behaviour reflects cognitive processing: fixation durations track processing effort, saccades relocate attention, and regressions, backward saccades to earlier text, are associated with comprehension difficulty [18, 19]. Eye-movement measures predict reading performance in on-line reading contexts [7], and recent work shows that global measures such as fixation counts and regression rates predict scores on standardised comprehension tests [12]. Salvucci and Goldberg’s taxonomy of fixation-identification algorithms [23] remains the reference point for converting raw gaze into oculomotor events; our platform implements a dwell-based, area-of-interest variant over live text tokens. This literature justifies gaze as the primary sensing channel for reading difficulty, but it does not by itself provide the infrastructure to act on the signal in real time.

2.2 Gaze-Contingent and Adaptive Reading Interfaces

Typography measurably affects reading for both general and impaired readers: letter spacing [3], font size [22], background colour [21], and reader-controlled text customisation [2] all shift performance

or comfort. Systems research has closed the loop between gaze and text since eyeBook’s gaze-responsive reading experiences and the reading assistance of Beymer et al. [4]. Gaze Tutor detected disengagement from gaze and triggered tutorial dialogue moves [5]; the Eye-Mind Reader detected mind wandering during reading and intervened with self-explanation prompts, improving long-term comprehension [9], and the approach has since been carried into real classrooms [8]. Within the Reading the Reader project, three predecessor prototypes demonstrated gaze-driven typographic adaptation for readers with central vision loss [14, 16, 20], and Jensen et al. introduced reading resume time as a measure of post-intervention disruption, finding that only a gradual intervention paired with highlighting significantly reduced recovery time [10]. Micro-intervention design frameworks likewise argue for small, just-in-time adaptations over wholesale changes [17]. Across this line of work the adaptive loop is repeatedly rebuilt: each system binds one sensor, one detection method, and one intervention repertoire into one interface, so intervention comparisons across systems remain confounded by the platforms themselves.

2.3 Experiment Tooling

General-purpose experiment software approaches the problem from the opposite side. Tobii Pro Lab [25] designs and records eye-tracking studies but does not drive real-time text adaptation; PsychoPy [15] and Psychtoolbox offer stimulus control and gaze recording with scripting, but adaptive reading interfaces, pluggable decision modules, and researcher-in-the-loop intervention control must be built from scratch. Consumer reading tools such as Immersive Reader [13] adapt presentation but sense nothing. To our knowledge, no available tool combines experiment design, gaze recording, real-time gaze-driven text intervention, and pluggable decision modules; this capability gap is what our platform targets.

2.4 Toward Low-Cost Sensing

Research-grade trackers such as the Tobii Pro Fusion [24] deliver the sampling rates and accuracy that reading research needs, but their cost and setup confine studies to the laboratory. Webcam-based eye tracking has been validated as a preliminary instrument for reading tests [11], and commodity cameras can additionally supply facial signals that research trackers do not observe at all. A platform whose sensing layer is a replaceable module can adopt such sensors as they mature without rebuilding the experiment infrastructure around them; Section 4 reports our work in progress in this direction.

3 THE PLATFORM

The platform is a two-screen, researcher-operated system. A participant reads reflowable Markdown-rendered text on a dedicated screen while a physical eye tracker streams gaze; on a second screen, a researcher console mirrors the participant’s view live, overlays gaze, fixations, and saccade paths, and exposes the experiment controls. Both surfaces are views of one authoritative, backend-owned session snapshot, so what the researcher sees is what the participant experiences (Figure 1).

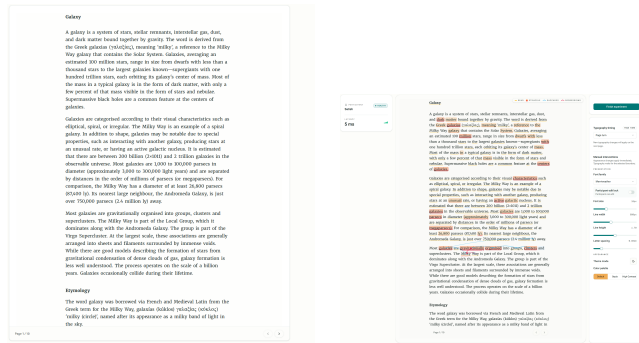


Figure 1: The participant’s reading view (left) and the researcher console mirroring the session live with gaze, fixation, and intervention controls (right).

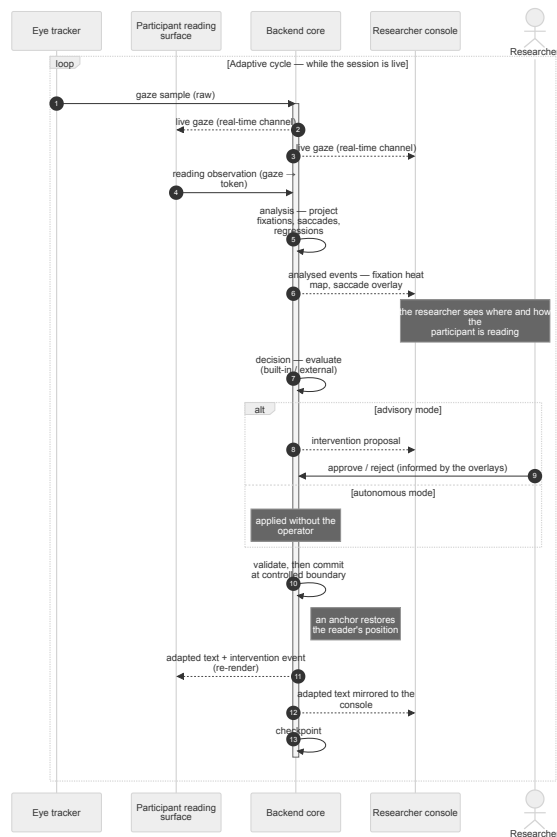


Figure 2: One pass through the adaptive loop: gaze flows from the eye tracker through analysis and decision to a context-preserving intervention commit, mirrored live on the researcher console; each stage is a replaceable module.

3.1 Four Modules Behind Stable Contracts

The adaptive loop is decomposed into four modules, each independently replaceable behind a stable contract (Figure 2):

Sensing produces gaze samples. The default adapter wraps a Tobii eye tracker through the vendor SDK, including nine-point calibration gated on a validation pass; a simulated source implements the same contract for development and demonstration without hardware.

Analysis projects gaze into oculomotor events: fixations, saccades, and regressions. A built-in dwell-based detector classifies fixations over text tokens in the spirit of area-of-interest identification [23]; alternatively, an external provider receives the gaze and reading observations and returns its own analysis.

Decision consumes the analysed state and returns at most one intervention proposal. Strategies range from a built-in rule-based policy through researcher manual action to external providers, including machine-learning pipelines.

Intervention validates and commits a typographic change: font family and size, line width and height, letter spacing, and colour theme, applied as presentation patches to the participant’s reader.

The backend follows a ports-and-adapters architecture whose dependency rule points strictly inward to a framework-free domain core. The rule is not a convention: it is enforced at build time and additionally checked by an executable boundary test that inspects the compiled dependency graph. Everything technology-specific, the Tobii SDK binding, the WebSocket transport, persistence, lives in adapters at the edge.

3.2 Two Channels and a Lock-Free Hot Path

Researcher commands travel over REST; high-frequency data travels over a dedicated WebSocket channel. This separation keeps the per-sample path free of request overheads and locks: gaze ingestion uses atomic operations only, while a single lifecycle gate serialises control-plane operations such as session start and stop. The latency budget for the full loop, from gaze sample to intervention dispatch, is 100 ms, and Section 5 reports the measured margins.

External module providers connect out of process over a documented protocol: a hello/welcome handshake declaring the module and its capabilities, heartbeats, and typed inbound/outbound envelopes. If a provider disconnects, the platform degrades gracefully to its built-in implementation and records the transition in the session record.

3.3 Mapping Gaze to Words in a Reflowable Text

Adaptive typography changes the very layout that gaze must be mapped onto, so the mapping cannot rely on static stimulus coordinates. The reader maintains live bounding boxes for every rendered word token and maps each gaze sample in two stages: line selection with hysteresis, then within-line word scoring. The hysteresis biases the mapping toward the currently attended line to absorb vertical jitter; because the mapping is computed against the current boxes, it survives reflow by construction and is independent of screen resolution. Dwell accumulated per token feeds the analysis module, and per-token attention renders live in the researcher console.

3.4 Context-Preserving Interventions

A typographic change that reflows the text can throw a reader to a different position, and prior work shows recovery is a measurable cost [10]. The platform separates a decision from its commit: an intervention decided now is committed at a controlled boundary, and the reader's context is carried across it. The runtime captures a position anchor every 120 ms, preferring the currently gazed word, falling back to the first visible word, then to the enclosing paragraph. At commit, a restore strategy repositions the view: appearance-only changes preserve the scroll offset, while layout-affecting changes re-anchor to the captured word. Crucially, the mechanism grades its own outcome: after every intervention it measures the residual distance between the restored anchor and its target and writes a graded outcome, preserved, degraded, or failed, with the pixel residual into the session record. This self-instrumentation is what later exposed a defect in our own restore strategy (Section 5.4).

3.5 Researcher Control and Reproducible Sessions

The researcher chooses among four execution modes: manual (researcher-applied interventions only), advisory (proposals await researcher approval), autonomous (proposals apply automatically), and control (no interventions). A guided setup flow gates session start on four readiness checks: participant, tracker, calibration, and reading material. During a session, every gaze sample, oculomotor event, proposal, decision, intervention, and context-restore outcome is appended to a single schema-versioned session record, checkpointed incrementally and sealed at session end. Records export as JSON or CSV and reconstruct fully in a replay mode with timed playback and scrubbing; every quantitative result in this paper was computed from exported records alone, not from live state.

4 WORK IN PROGRESS: WEBCAM SENSING BEHIND THE SAME CONTRACT

Research-grade eye trackers anchor the platform's primary signal, but they also anchor its deployments to a laboratory. Commodity webcams invert both properties: they are already attached to nearly every reading device, and they observe a channel the eye tracker cannot, the reader's face. As a first exercise of the sensing boundary with a signal quite unlike gaze, we have implemented a webcam-based facial-state module. We describe it here as work in progress: it is integrated end to end, but it has not yet been evaluated with participants, and we state its present limits explicitly.

4.1 A Facial-State Module

The module treats the webcam as a supplementary context source rather than a replacement eye tracker. The platform tracks its gaze source and its face source independently, and exposes four sensing configurations: eye tracker only, webcam only, a hybrid mode in which the Tobii tracker remains the authoritative gaze source while the webcam contributes facial state, and a simulated mode for development.

The built-in implementation is a background worker that samples the camera at a configurable interval (200 ms by default) and processes each frame classically, without a learned model: face

detection via a Haar cascade, followed by a 68-point facial landmark fit. From the landmarks it derives interpretable, geometry-only features: per-eye openness, blink likelihood, mouth tension, head offset from the face-box centre, frame-to-frame motion, and a capture-quality score from image sharpness. A rule-based classifier folds these into a coarse three-state reading-difficulty signal, neutral, possible struggle, or possible ease, using fixed thresholds on blink likelihood, motion, and mouth tension. In webcam-only mode the worker additionally emits a coarse gaze proxy from a per-eye dark-pixel pupil estimate; this proxy is uncalibrated and is not a point-of-gaze estimator, so webcam-only mode does not support word-level reading analysis.

Everything the module produces flows through the same paths as gaze: facial observations, difficulty signals, and sensor status events are broadcast live, written into the session record, and reconstructed by replay. The worker degrades gracefully, reporting distinct degraded and unavailable statuses when the camera cannot be opened, no face is found, or the landmark fit fails, consistent with the platform's behaviour when an external provider disconnects.

4.2 Replaceability by Construction

The facial-state seam mirrors the analysis and decision seams: an out-of-process provider can register for the facial-state module over the same provider protocol and submit observations, status, and gaze samples. When an external provider is active, the built-in worker pauses itself. The intended trajectory is therefore not to perfect the built-in heuristic but to replace it: a learned facial-analysis pipeline, or a validated webcam gaze estimator in the spirit of Lin et al. [11], can take over the module without any change to the session runtime, exactly as an independent team's analysis pipeline replaced our built-in fixation detector in the evaluation (Section 5.3).

4.3 Present Status and Limits

We report the module's maturity precisely. The capture, feature extraction, classification, recording, and replay path is implemented and runs; the sensing-mode gating (including skipping tracker calibration in webcam mode) is part of the guided setup flow. However, the difficulty heuristic is hand-tuned and unvalidated, no participant sessions have yet been recorded with the webcam active, the researcher console does not yet render the facial-state stream in its live view, and the gaze proxy is far below the accuracy that reading analysis requires. The near-term configuration we consider most promising is the hybrid mode, tracker gaze plus webcam facial context, feeding decision strategies with a signal dimension that gaze alone lacks; validating the facial-state signal against ground truth during real reading sessions is the immediate next step (Section 7).

5 EVALUATION

The evaluation asks three questions, matching the platform's claims. **EQ1 (Feasibility)**: does the platform sustain research-grade sensing and stay within its latency budget on real hardware? **EQ2 (Modularity)**: can modules genuinely be replaced by outside implementations without modifying the core? **EQ3 (Context preservation)**: does the intervention mechanism preserve the reader's context, and what does its self-instrumentation reveal? We deliberately do

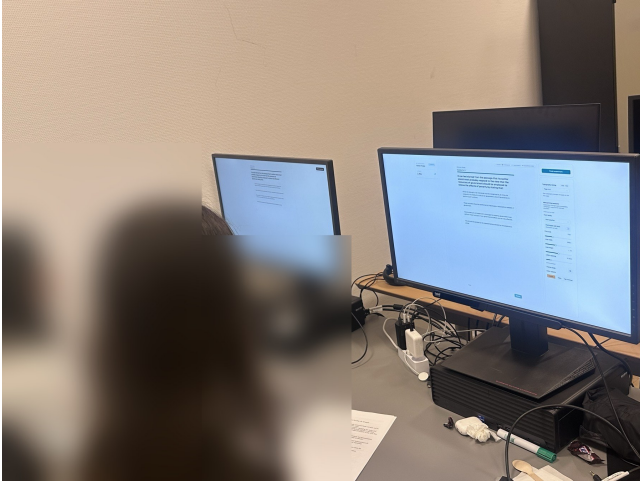


Figure 3: The two-screen evaluation rig: the participant reads on the eye-tracked display while the researcher operates the console on the second screen.

not claim intervention efficacy; with four participants, all behavioural comparisons below are descriptive.

5.1 Setup

Sessions ran on a single host: a Tobii eye tracker (model IS4 Large Peripheral, rated 90 Hz) below a 2560×1440 participant display, with the researcher console on a second screen and all network traffic over loopback (Figure 3). Four participants each read an expository Markdown article under two conditions, with and without context-preserving restore, yielding eight sessions; interventions were applied manually by the researcher. Every session passed a nine-point calibration with validation before reading began. A separate 91-second advisory-mode run exercised the automated decision path for latency measurement. Participants were drawn from the project team and acquaintances; sessions produced 186–475 s of reading and 11,063–42,683 gaze samples each. [Insert the approved consent and data-handling statement for these sessions before circulation.]

5.2 EQ1: Real-Time Feasibility

Sensing quality. Across the eight sessions the platform sustained a mean sampling rate of 89.9 Hz (SD 0.2 Hz) against the tracker’s rated 90 Hz, with every session within 0.6 Hz of nominal and inter-sample intervals clustering at the nominal 11.1 ms period. Gaze validity during naturalistic reading ranged from 92.5% to 98.0% (mean 95.6%). Calibration accuracy averaged 0.30–0.69° with precision 0.08–0.33° across participants.

Latency. The end-to-end budget is 100 ms from gaze sample to intervention dispatch. Client round trips stayed far inside it (Figure 4): participant-side median 1 ms with a 95th percentile of 3 ms, researcher-side median 1 ms with a 95th percentile of 8 ms, and no measured sample in either stream exceeded the budget. On the advisory-mode run, the in-process decision pipeline, from freshest gaze sample to proposal dispatch, showed a median of 6 ms, a 95th

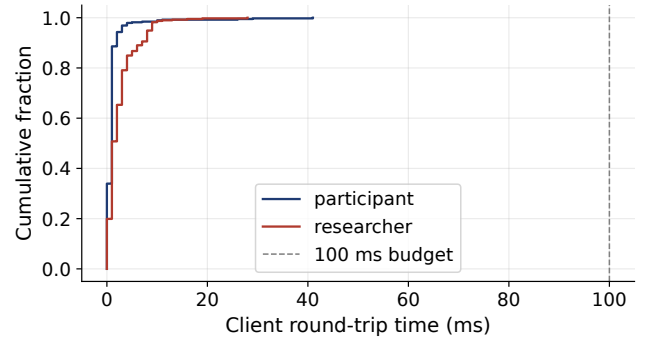


Figure 4: Distribution of client round-trip times against the 100 ms budget; no measured sample exceeded it.

percentile of 11 ms, and a maximum of 46 ms over 14,015 samples; round trips to the out-of-process decision provider had a median of 5 ms and a maximum of 7 ms over its 8 proposals. Real-time adaptation therefore leaves most of the budget as headroom, even with decisions delegated across a process boundary.

5.3 EQ2: Modularity

The dependency rule is enforced at build time and verified by an executable boundary test over the compiled dependency graph, so the claimed module boundaries cannot silently erode. The sharper test of a contract, however, is whether someone who did not write it can implement it. Three external providers connected to the platform over the provider protocol: our reference decision provider, our reference eye-movement analyser, and, critically, a reading-difficulty pipeline built by an independent team (detecting cognitive load and reading style from gaze) that was integrated through a thin connector without modifying either codebase. That third-party pipeline did not merely connect: it served as the live analysis source for all eight evaluation sessions, meaning the fixation, saccade, and regression events in our own evaluation data were produced by an outside team’s algorithm running against the published contract. When a provider disconnects mid-session, the platform falls back to its built-in implementation and logs the degradation in the session record.

5.4 EQ3: Context Preservation

Live sessions, descriptively. The four preservation-enabled sessions contained 19 interventions; the four disabled sessions contained 26. Median reading-resume time after an intervention was 482 ms with preservation versus 650 ms without (means 670 versus 923 ms), and the post-intervention regression rate within a 5 s window was 23% with preservation (down from a 29% pre-intervention baseline) versus 33% without (baseline 32%). The direction was stable across 3, 5, and 10 s windows and for three of four participants. We repeat: $n=4$, no inferential statistics, and the preservation condition bundles the positional restore with a visual highlight cue, a confound we return to in Section 6.

What self-instrumentation revealed. The mechanism’s own outcome grading contradicted the behavioural trend: of the 19 preservation enabled interventions, only 3 graded as preserved and 16 as degraded, with a median residual of 389 px against a median induced displacement of only 38 px. The original restore strategy was over-correcting, moving the reader roughly an order of magnitude further than the reflow it compensated. We would not have found this from the behavioural measures; the platform’s insistence on measuring its own disruption is what exposed it.

Deterministic sweep and the revised restore. To characterise the defect and verify a fix, we ran a scripted sweep, 11 layout-affecting interventions crossed with 6 reading positions, 66 trials per condition, in a browser-automation harness measuring the true displacement of the reader’s word. Without preservation, median displacement was 34.3 px, but heavy-tailed (large font steps reached 142–188 px). With the original restore, the median rose to 104.4 px, three times worse than doing nothing, and 45 of 66 trials over-repositioned, including a letter-spacing change with zero natural displacement being thrown 115 px. A revised restore, which holds the captured reading offset instead of snapping the anchor to the viewport top, brought the median to 32.4 px, at or below the no-preservation baseline, with only 4 of 66 trials over-repositioning and 61 of 66 grading as preserved (median residual 31 px), as Figure 5 shows. One reading line at evaluation typography is 32.4 px: the revised restore holds the reader within about one line even under the largest reflows. The revised strategy is verified geometrically; behavioural re-validation with readers is future work.

5.5 Functional Coverage and Reproducibility

The backend’s automated suite (100 tests) passes, and the build-enforced boundary test runs with it. [Re-run dotnet test before submission and confirm the exact test count.] All figures and tables in this section were reconstructed from exported session records alone, using the platform’s replay path rather than any live-state logging, which exercises the reproducibility claim in practice: the session record is sufficient to re-derive the evaluation.

6 DISCUSSION

6.1 What the Platform Enables

The comparative questions that motivated this work become configuration rather than engineering. A study in the style of Jensen et al. [10], comparing intervention designs on recovery cost, needs no new system: interventions, decision strategies, and analysis providers are selected per session, and every trial leaves a replayable record. Because decision strategies sit behind a process boundary, machine-learning teams can study their models against live readers without touching, or even reading, the platform’s core, as the independent reading-difficulty pipeline in our evaluation demonstrated. The same seam positions the platform for AI-driven decision strategies, which can be compared against rule-based and human strategies under otherwise identical conditions.

6.2 Measure the Cost of the Quality You Claim

The over-repositioning finding generalises beyond this system. Our restore strategy was plausible, implemented carefully, and wrong:

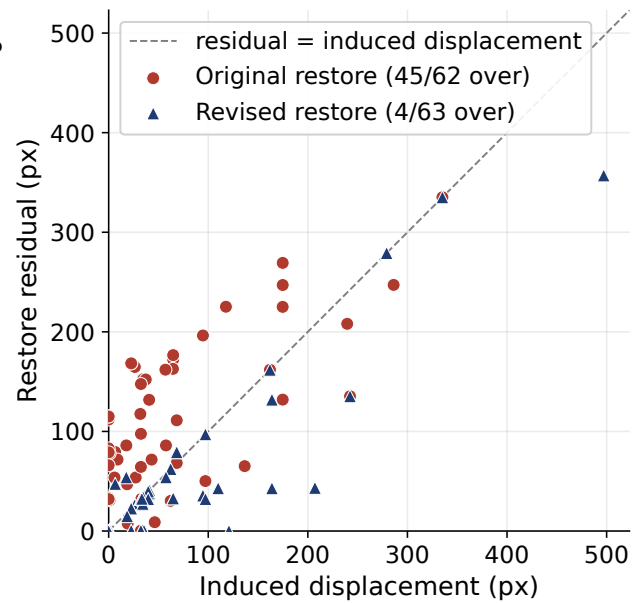


Figure 5: Word displacement across the 66-trial deterministic sweep: the original restore over-repositions relative to no preservation, while the revised restore holds displacement at or below the baseline. [The embedded legend counts trials with measurable residuals (45/62 and 4/63); the text reports over-repositioned trials out of all 66. Reconcile the denominators against Experiments/analysis before submission.]

it moved readers three times further than the disturbance it compensated, and the behavioural measures alone would not have told us, because the bundled highlight cue plausibly masked the geometric damage. The defect surfaced only because the mechanism grades every restore it performs and writes the residual into the record. We suggest this as a design principle for adaptive interfaces generally: a system that claims a quality attribute, context preservation, smoothness, unobtrusiveness, should instrument the cost of that attribute in production data rather than asserting it. Two further principles from this work: enforce module boundaries in the build rather than in convention, and treat an outsider’s independent implementation, not the authors’ own reference implementations, as the test that a contract is truly assumption-free.

6.3 Limitations

We state the limits plainly. The behavioural comparison covers four participants, one text, and manual intervention only; the participants included members of the project team; nothing here is an efficacy claim. The with-preservation condition bundles positional restore with a highlight cue, so the behavioural gap cannot be attributed to geometry alone, particularly since prior work suggests highlighting itself drives recovery [10]. The revised restore is verified over a deterministic geometric sweep, not yet with readers. The automated decision path was validated in a single advisory run rather than a campaign. The analysis events in the evaluation were produced by the external provider’s algorithms, which have

not been calibrated against published oculomotor norms. The webcam facial-state module is integrated but unvalidated, and its difficulty heuristic is hand-tuned. Finally, the frontend is verified by manual walkthrough rather than an automated suite, and the platform currently renders Markdown content only.

7 CONCLUSION AND FUTURE WORK

We presented a researcher-operated platform for real-time adaptive reading experiments in which sensing, analysis, decision, and intervention are independently replaceable modules behind stable, build-enforced contracts. On real eye-tracking hardware it sustained research-grade sensing with wide latency headroom, its analysis was performed live by an independent team's pipeline through the published provider contract, and its self-instrumented intervention mechanism exposed and then verified the correction of a threefold over-repositioning defect in our own restore strategy. A work-in-progress webcam module extends the sensing boundary toward commodity hardware.

Future work follows directly from the stated limits. First, a behavioural re-validation of the revised restore that crosses positional restoration with the highlight cue, unbundling the confound the present data cannot resolve. Second, validation of the webcam facial-state signal against ground truth during real reading sessions, surfacing it in the researcher console, and replacing the built-in heuristic with a learned provider, including a calibrated webcam gaze estimator [11]. Third, a campaign-scale study in autonomous mode with a learned decision strategy, moving the automated path from single-run validation to routine operation. Fourth, replacing the hand-tuned line-selection hysteresis with a principled uncertainty model such as a Kalman filter. The platform, its documentation, and the anonymised session records are available as open source, and we invite external analysis, decision, and sensing providers to test their implementations, and our contracts, against it. **[Insert the public artifact URL after anonymising the session data files, which currently carry participant first names.]**

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